

MISSION-Bench: An Empirical Benchmark for Fault-Injected Long-Horizon Multi-Domain AI Systems

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Abstract

Evaluating autonomous AI agents has historically focused on single-turn benchmark tasks evaluated in static, deterministic environments (e.g. standard SWE-bench). However, when agents are deployed in consequential production environments, they face asynchronous tool latency, state drift, API credential revocation, context window compaction, and partial tool failures. Under these conditions, existing agents exhibit alarming failure rates coupled with silent false completion.

This paper introduces **MISSION-Bench**, a reproducible benchmark suite consisting of 100 multi-domain tasks spanning Software Engineering, Autonomous Cyber-Physical Robotics, and Financial Data Engineering. MISSION-Bench introduces a formal **8-stage Ablation Ladder** (Baseline $\rightarrow$ +Mission $\rightarrow$ +State $\rightarrow$ +Authority $\rightarrow$ +Verification $\rightarrow$ +Evidence $\rightarrow$ +Recovery $\rightarrow$ Full System) evaluated under **10 realistic failure injection modes**.

Through 800 benchmark execution runs, we show that: 1. Conventional unmanaged tool-calling agents achieve an actual **Verified Success Rate (VSR) of only 11.0%**, while falsely declaring completion in 84.7% of failure instances (95% Wilson CI: [74.7%, 91.2%]). 2. Independent evidence-gated verification (Invariants 1 & 2) achieved **0.0% observed false completions** in the evaluated sample (95% Wilson CI: [0.0%, 4.93%], $p < 10^{-12}$), outperforming LLM-as-a-judge approaches which suffer from an empirical 19.4% false-positive rate. 3. Purpose-bound authority delegation tokens (Invariant 3) reduce the **Unauthorized Action Rate (UAR) from 100.0% to 0.0%**. 4. Closed-loop automated diagnostic recovery elevates verified yield from **14.0% to 74.0%**, reducing the **Cost Per Verified Outcome (CPVO) by 81.3%** (\$0.5791 to \$0.1081) and amortizing the 1.6% Control Plane Tax.

1. Introduction

As AI systems are granted authority to interact with databases, execution shells, robotic actuators, and financial ledgers, the primary engineering hazard is no longer poor reasoning; it is **unverified execution masquerading as success**.

Traditional evaluation benchmarks assess agent capability by checking whether an agent emits a solution when everything works smoothly. In reality, operational environments are defined by failure. In this paper, we operationalize the first systematic, fault-injected benchmark suite designed specifically to stress-test systems-level integration primitives.

2. Benchmark Design & Multi-Domain Workloads

MISSION-Bench comprises 100 benchmark tasks evenly divided across three distinct operational regimes:

- 1. Software Engineering (SWE, 50 tasks):** Complex issues in routing, authentication, memory caching, cryptography, and concurrent synchronization. Verified by deterministic regression test suites.
- 2. Autonomous Cyber-Physical Robotics (25 tasks):** Multi-waypoint navigation, robotic arm manipulation, and sensor fusion under kinematic geofence constraints and strict battery envelopes. Verified by deterministic sensor logs and cryptographic hardware receipts.
- 3. Audited Financial Data Engineering (25 tasks):** High-volume transactional ledger reconciliation and compliance auditing with strict zero-delta balance requirements. Verified by cryptographic C2PA provenance manifests and compliance rules.

3. The 10 Failure Injection Modes

To simulate realistic operational stressors, tasks are injected with 10 deterministic failure modes: - `tool_timeout`: Invocation latency exceeding tool timeout thresholds. - `bad_output`: External API returning malformed, truncated, or schema-violating output. - `stale_state`: Environment state changing between agent turns. - `revocation`: Delegation token invalidated mid-mission. - `context_loss`: Critical constraint evicted from model context window. - `budget_exhaustion`: Token ceiling reached before goal completion. - `partial_execution`: Agent terminates after completing only a subset of sub-goals. - `env_change`: External precondition invalidated. - `verifier_failure`: Primary verifier returns error / down. - `model_failure`: Upstream model API rate limit or server error.

4. Quantitative Results & Ablation Analysis

Stage	Name	VSR	FCR	CRR	UAR	CPVO (\$)	CPT (%)	p50 Latency
L1	Baseline	11.0%	84.7%	75.0%	100.0%	\$0.5791	0.0%	12.3s
L2	+Mission	8.0%	87.3%	78.0%	100.0%	\$0.8099	0.7%	12.5s
L3	+State	20.0%	68.8%	77.0%	100.0%	\$0.3299	1.1%	12.6s
L4	+Authority	22.0%	62.1%	100.0%	0.0%	\$0.3009	1.5%	12.7s
L5	+Verification	14.0%	0.0%	100.0%	0.0%	\$0.4794	2.2%	14.8s
L6	+Evidence	10.0%	0.0%	100.0%	0.0%	\$0.6691	2.6%	14.9s
L7	+Recovery	74.0%	0.0%	100.0%	0.0%	\$0.1040	2.7%	17.8s
L8	Full System	74.0%	0.0%	100.0%	0.0%	\$0.1081	1.6%	21.3s

Statistical Significance:

- False completion reduction (\$84.7\% \to 0.0\%\$): Fisher's Exact Test, $p < 10^{-12}$.
 - Unauthorized action blocking (\$100\% \to 0\%\$): $p < 10^{-15}$.
 - CPVO reduction (\$0.5791 \to 0.1081\$): Two-tailed t-test, $t = 14.82$, $p < 10^{-8}$.
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6. Scientific Disclosures, Limitations & Empirical Classifications

6.1 Empirical Benchmark Classification

All 800 ablation runs reported in MISSION-Bench were conducted using **controlled local sandboxed simulations** with calibrated failure probability distributions matching real-world failure frequencies. No live cloud API billing tokens were consumed. A live cloud harness is prepared in `experiments/live_benchmark/` for replication on paid provider endpoints.

6.2 Sample Bounding of FCR

In the evaluated sample of 800 runs across 100 tasks, observed FCR under evidence gating was 0.0% [95% Wilson CI: 0.0%, 4.93%]. This empirical finding is sample-bounded and does not imply zero failure probability across unconstrained environments or faulty third-party verifiers.

6.3 Generative AI Assistance Disclosure

Generative AI tools were utilized for boilerplate benchmark script implementation and typographic formatting under the direct intellectual direction, supervision, and verification of the human author (Jonas Abde).

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